Project Proposal

The "Gut Check": Profiling Gut Microbial Diversity in Amyotrophic Lateral Sclerosis

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May 20, 2026

1 Background

Amyotrophic lateral sclerosis (ALS) is a fatal motor neuron disease. The motor neurons in the brain and spinal cord progressively die, and as they do, patients lose control of their muscles. They can no longer walk, then they cannot swallow or speak well, and eventually they cannot breathe on their own (Brown and Al-Chalabi 2017). Roughly 90% of cases are sporadic, meaning the patient has no family history of ALS and no single mutation can be blamed. Genes do contribute to risk, but no clear environmental trigger has been found for sporadic ALS, even after many years of work.

One thing researchers have started paying more attention to is the gut microbiota. The gut talks to the brain through three main routes: the vagus nerve, the immune system, and bacterial metabolites that travel through the bloodstream. This back-and-forth is called the gut-brain axis (Cryan et al. 2019). When the bacterial community shifts away from a healthy state, a condition called dysbiosis, the signals along all three routes get disturbed. Dysbiosis has been linked to inflammatory bowel disease, depression, and Parkinson's disease, among others.

The reason to look at the microbiome in ALS specifically comes mostly from a 2019 paper by Blacher and colleagues. Using the SOD1 mouse model, they showed that wiping out the gut microbiota with antibiotics made the disease worse, while giving the mice certain bacteria such as *Akkermansia muciniphila* slowed it down. They traced the protective effect to a bacterial metabolite called nicotinamide and found lower cerebrospinal-fluid nicotinamide in a small group of human ALS patients (Blacher et al. 2019). Several 16S surveys in human patients have come out since then, reporting differences in alpha diversity and in major phyla, but the results across cohorts do not always agree.

For this project I will reanalyze one of those human cohorts to see whether the gut bacterial community of ALS patients differs from that of healthy controls.

2 Biological Question

Does the gut bacterial community in ALS patients differ from that of healthy spouse controls, both in within-sample diversity (alpha) and in between-sample composition (beta)?

3 Hypothesis

I expect ALS patients to show:

(i) lower Shannon alpha diversity than their matched controls, and
(ii) visible separation from controls in a Bray-Curtis PCoA ordination.

If a separation is present, I expect it to be driven by shifts in specific bacterial genera that other dysbiosis studies have flagged.

4 Dataset

The data come from Hertzberg and colleagues (2021) (Hertzberg et al. 2021) and are public on NCBI as BioProject PRJNA566436. There are 18 stool samples in total: 9 ALS patients and their 9 spouses as matched controls, giving 9 household pairs. The V4 region of the 16S rRNA gene was amplified with the 515F and 806R primers and sequenced on an Illumina MiSeq.

I picked this cohort mainly because of the spouse-paired design. Spouses tend to eat the same food, drink the same water, share a household, and have done so for years. That gets rid of a lot of the confounding that usually makes microbiome studies hard to interpret, so a difference between case and control is more likely to be about the disease and not about lifestyle.

The raw reads have already been run through DADA2 (Callahan et al. 2016) against the SILVA reference database to produce an ASV count table and a per-ASV taxonomy table. My analysis starts from those two files, as the assignment says.

5 Methods

The analysis will run in Python in a single Jupyter notebook. Table 1 lists the eight steps in execution order and the libraries used at each.

Table 1: Python tools used at each step of the pipeline.

Step	What it does	Python tool
1	Load the ASV count table, the taxonomy, and the sample metadata	`pandas`
2	Drop rare ASVs (present in fewer than 5% of samples, with a floor of two samples for this small cohort)	`pandas`
3	Convert counts to per-sample relative abundances	`pandas`
4	Compute Shannon alpha diversity per sample and compare ALS to control with a paired Wilcoxon signed-rank test	`scikit-bio`, `scipy.stats`
5	Compute the Bray-Curtis beta diversity matrix	`scikit-bio`
6	Project the distance matrix to two dimensions with PCoA and plot it	`scikit-bio`, `matplotlib`, `seaborn`
7	Test group separation with PERMANOVA on the distance matrix	`scikit-bio`

Step	What it does	Python tool
8	Identify driver genera with per-genus Mann-Whitney U tests and Benjamini-Hochberg false discovery rate correction	`scipy.stats`, `statsmodels`

The full software stack is `pandas`, `numpy`, `scipy`, `scikit-bio`, `statsmodels`, `matplotlib`, `seaborn`, and `jupyter`.

6 Expected Outcomes

By the end of the project I will have:

1. A PCoA scatter plot of the Bray-Curtis distances, with each sample colored by disease status. I expect ALS and control samples to at least partially separate.
2. A boxplot of Shannon diversity by group, plus the result of the paired Wilcoxon test on the household pairs.
3. A short ranked list of bacterial genera whose abundance differs significantly between ALS and control samples after FDR correction.
4. A reproducible Jupyter notebook that performs every step above and rebuilds every figure from scratch.

7 Limitations

There are a few things worth flagging before I start. First, 9 pairs is a small sample. Only fairly big effects will turn up as statistically significant, and a null result will not really mean much. Second, 16S can usually only resolve bacteria to the genus level, so I cannot say much about specific species. Third, this is an observational study, so even a strong association between microbial composition and ALS would not prove cause and effect on its own. I will talk about all of this in the final report.

References

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